

Bintang Dwi Marthen

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Research Interests

Systems-for-ML, ML-for-systems, operating systems, cloud computing, storage systems, and distributed systems.

Education

Bandung Institute of Technology

MS in Computer Science

Bandung, Indonesia
Sep 2025 – Exp. Jul 2026

- #1 Engineering School in Indonesia ([article](#) 🔗)
- GPA: 3.85/4.0 ([study report](#) 🔗)
- **Coursework:** Advanced Distributed Systems, Advanced Operating System, Cloud Computing Architecture, Deep Learning

Bandung Institute of Technology

BS in Computer Science

Bandung, Indonesia
Aug 2021 – Jul 2025

- #1 Engineering School in Indonesia ([article](#) 🔗)
- GPA: 3.89/4.0 (Major GPA: 3.86/4.0) ([transcript](#) 🔗)
- **Coursework:** Computer Organization and Architecture, Operating System, Computer Networks, Parallel and Distributed Systems, Distributed Application Architecture, Machine Learning, Database Management

Research Experience

Serverless Platform for Development Workload

International Research Collaboration

Harvard University
Apr 2025 – Present

- Collaborating with Assistant Professor Juncheng Yang (Harvard University) on designing a developer-friendly serverless platform to eliminate idle resource waste inherent in VM-based development workflows.
- Analyzed fine-grained execution traces from thousands of workloads to systematically characterize latency and resource inefficiencies in emerging serverless environments.
- Discovered that container image construction dominates startup overhead, accounting for up to 82% of total initialization time.
- Investigating optimized base images synthesis and caching strategies to reduce initialization overheads and enable low-latency, high-reuse execution environments.

Feasibility of Static Parameterization in Adaptive Caches

International Research Collaboration

Harvard University
Mar 2025 – Present

- Collaborating with Assistant Professor Juncheng Yang (Harvard University) on re-evaluating how much adaptivity is actually needed in cache management.
- Demonstrated through extensive simulation that statically parameterized cache policies can outperform their adaptive counterparts by up to 20% in miss ratio.
- Uncovered that adaptive cache performance depends on the granularity match between the adaptation mechanism and the workload's phase behavior; mismatches in either direction lead to performance loss.
- Investigating phase-change and workload drift detection mechanisms to enable coarse-grained, stability-aware adaptivity with minimal runtime and tuning overhead.

System and AI Research (SYAIR) Training Program

Research Training Program

University of Chicago
Jan 2025 – Jul 2025

- Selected as one of the top 50 students in Indonesia for a research training program led by Prof. Haryadi S. Gunawi (University of Chicago).
- Reproduced the *Clock-PRO* cache replacement algorithm as a Chameleon Trovi artifact.
- Studied over 30 research papers from top systems conferences (*SOSP*, *OSDI*, *NSDI*, *FAST*, and others).

Sentiment Analysis Using LLMs for Twitter Data

Undergraduate Researcher

Bandung, Indonesia

Oct 2024 – Dec 2024

- Collaborating with Ayu Purwarianti (Bandung Institute of Technology) and Achmad Imam Kistijantoro (Bandung Institute of Technology) on leveraging large language models for sentiment analysis of Indonesian Twitter posts.
- Fine-tuned pre-trained transformer models and evaluated domain adaptation strategies for low-resource Indonesian text.
- Maintained and optimized NVIDIA DGX servers for large-scale model training.
- Built an internal web interface to streamline prompt testing and result comparison across multiple LLMs.

Work Experience

Research Intern

Formulatrix

Bandung, Indonesia

Jun 2024 – Sep 2024

- Designed and implemented a proof of concept for the automatic map discovery feature of the ROVER product, demonstrating feasibility and informing future production development
- Introduced Simultaneous Localization and Mapping (SLAM) using a monocular front-facing camera, validating navigation capability in lab environments
- Modified OpenCV ArUco to support STag markers in singular and composite configurations, improving marker flexibility for prototype testing

Highlighted Projects

Simple x86 32-bit Operating System

[github repo](#) 

- Developed a basic operating system kernel from scratch, with support for interrupts, keyboard input, and FAT32 file system management
- Implemented key file system operations such as `ls`, `cd`, `mkdir`, `whereis`, `cat`, `rm`, `cp`, and `mv`, providing full CRUD functionality
- Developed an electronic classroom where multiple users can simultaneously view and draw on a "chalkboard" with each person's edits synchronized

Clock-PRO Cache and CACHEUS Fix

[github repo](#) 

- Implemented Clock-PRO cache algorithm on libCacheSim, a high-performance library for cache simulation
- Debugged and fixed faulty CACHEUS implementation

TCP From UDP

[github repo](#) 

- Developed a TCP-like connection over UDP socket using Python programming language
- Developed a file transfer protocol from the TCP connection
- Developed a tic-tac-toe game over the TCP connection

Technical Skills

Languages: C++, C, Python, Go, C#, SQL, JavaScript, Kotlin, Java

Operating Systems: Windows, WSL, Linux

Machine Learning: vLLM, Ollama, Transformers, PyTorch, TensorFlow, Scikit-learn, Pandas

Databases and Systems: SQLite, MySQL, PostgreSQL, MongoDB, Redis, CouchDB

Systems: RabbitMQ, FEMU, libCacheSim

Cloud Computing: Google Cloud, AWS, Chameleon Cloud, CloudLab, Amazon S3

DevOps: Kubernetes, GitHub CI/CD, containerd, firecracker microVM, Docker

Misc: LaTeX, Make, CMake

References

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